

Design Verification Plan and Report (virtual test version) - VBF-T3R2-DVPR-01

Case t3_r2 - power-tool battery. Every result value is mechanically taken from simulation output files; uncovered conditions are honestly listed as N/A. Model: DFN, lumped thermal, plating option enabled on the charge protocol.

Verification items

#	Item	Condition	Result value	Determination (vs entry-0 criteria)	Source
1	1C discharge capacity	1C CC discharge to 2.5 V, 25 °C ambient	3.3278 Ah	PASS (≥ 2 Ah)	cell/r2_p1_1c_dfn.json:capacity_ah
2	5C discharge capacity retention	5C CC discharge to 2.5 V, 25 °C, 720 s protocol	98.89 % (5C 3.2911 Ah $\div$ 1C 3.3278 Ah, same parameters incl. re-based nominal)	PASS (≥ 95 %)	cell/r2_p1_derived.json:capacity_retention_5c
3	5C discharge peak temperature	same run as #2	322.22 K (49.07 °C)	PASS (≤ 333.15 K)	cell/r2_p1_derived.json:t_max_5c_k
4	4C fast-charge lithium plating	4C CC charge to 4.2 V at 45 °C ambient (after 1C depletion), plating option on	anode surface potential min +0.0364 V - never below 0 V -> not plated	PASS (plated == false)	cell/r2_p1_4c_dfn.json:anode_potential_v
5	4C fast-charge peak temperature	same run as #4	332.52 K (59.37 °C)	PASS (≤ 333.15 K)	cell/r2_p1_4c_dfn.json:T_max_K
6	Power density	contract formula $V_{OC}^2 / (4 \cdot DCR) / \text{mass}$; $DCR = (V_{start} - V_{@10\%}) / I_{1C}$	63,130 W/kg	PASS (≥ 4000 W/kg)	cell/r2_p1_energy.json:power_density_w_kg
7	Voltage window	parameter set limits	2.5 - 4.2 V	PASS (design datum)	Chen2020 parameter set
8	Rated energy / energy density	1C discharge energy integral; mass per contract	12.13 Wh; 407.6 Wh/kg; 837.4 Wh/L	Informative (no contractual threshold)	cell/r2_p1_energy.json

Items explicitly not covered in this task

Item	Status	Note
Nail penetration	N/A (beyond pure-simulation boundary, requires physical experiment)	abuse scenario not in task contract
Overcharge to thermal runaway	N/A (protocol exists in toolchain; not required by task contract and not run - honest, not fabricated)	-
Crush / drop	N/A (requires physical experiment)	-
Cycle life	Not simulated (aging-capable parameter set available; task contract has no cycle requirement)	-
Rate-pulse DC resistance	Not simulated; the DCR datum is the calc-energy start-to-10% formula	-

Conclusion

All six contractual verification items (capacity, 5C retention, 5C T_max, 4C plating, 4C T_max, power density) PASS on the delivered design P1_power_stack (log round 2, verdict=pass by bda log-evaluate). Uncovered items for the paper's limitations section: physical abuse tests, cycle life, pulse resistance, and pack-level thermal management (the 4C T_max

margin is 0.63 K under $h = 10 \text{ W/m}^2\text{K}$ - pack cooling must be confirmed at system level).